

AIGR Gene Hypothesis Deep Research —

Dictyostelium discoideum statC (Q54BD4)

Focus: function_assignment · **Hypothesis slug:** proliferation-and-defense-response **Target:** statC / Dd-STATc / dstC (UniProt Q54BD4; dictyBase DDB_G0293532; 931 aa) **Candidate GO terms under adjudication:** GO:0042127 (regulation of cell population proliferation), GO:0006952 (defense response)

Summary

Executive verdict: over-annotated / refuted for both candidate terms. Neither GO:0042127 (regulation of cell population proliferation) nor GO:0006952 (defense response) is supported by *Dictyostelium* experimental evidence for statC. Both terms are attached to statC exclusively by phylogenetically-inferred annotation (IBA, ECO:0000318, GO_REF:0000033) propagated from the metazoan STAT clade through PANTHER node PTN000927860. The "with/from" evidence sets for these IBA calls point to animal STATs (human STAT1 P42224, STAT2 P52630, STAT5A P42229) and — critically for the proliferation term — to the *Dictyostelium* paralog statB/dstB, **not** to any experimental result on statC itself.

When the two terms are adjudicated separately, as the seed hypothesis demands, they fail for different reasons. **GO:0042127 is a paralog carry-over:** the only experimentally supported proliferation phenotype among the three *Dictyostelium* STATs belongs to statB/dstB (Q70GP4), whose null cells are gradually outcompeted in co-culture with wild-type ([PMID:14701681](#), IMP). statC and statA carry the same term only by IBA. **GO:0006952 has no experimental basis in any *Dictyostelium* STAT:** genuine amoebal anti-bacterial defense is executed by sentinel (S) cells via the TIR-domain protein TirA and ROS-generating NADPH oxidases ([PMID:26927887](#); [PMID:17673666](#)), a machinery in which statC is not implicated. STATc's only documented relationship to oxidative conditions is as a stress-activated transcription factor (GO:0006979, IDA), not as an effector of pathogen defense.

By contrast, statC's experimentally supported core function is clear and well-documented: it is a **stress-activated, DNA-binding transcription factor** governing the transcriptional response to hyperosmotic, heat, oxidative, and DIF-1 (prestalk-inducer) signals, and it directs prestalk

cell differentiation and sorocarp morphogenesis during development. Activation is non-canonical — driven by inhibition of the tyrosine phosphatase PTP3 rather than by the metazoan JAK/cytokine-receptor cascade. Because the interferon/cytokine-driven proliferation and immune-defense functions of animal STATs depend on exactly that receptor system, which *Dictyostelium* lacks, the phylogenetic transfer of these two functions to statC is mechanistically unjustifiable. The recommended curation action is to remove or de-prioritize (mark non-core / NOT) GO:0042127 and GO:0006952 on statC, retaining the well-supported stress-response and developmental transcription-factor annotations.

Key Findings

F001 — Both candidate terms are IBA-only carry-overs from animal STATs with zero *Dictyostelium* experimental support

QuickGO retrieval for Q54BD4 (39 annotations) shows that **GO:0042127 (regulation of cell population proliferation) is annotated solely by IBA** (ECO:0000318, GO_REF:0000033, PANTHER PTN000927860), with a with/from set consisting of animal STATs including human STAT1 (P42224) and STAT5A (P42229). Likewise, **GO:0006952 (defense response) is annotated solely by IBA** (same PANTHER node), with a with/from set of human STAT1 (P42224) and STAT2 (P52630). Neither term carries any IDA, IMP, IEP, or HDA experimental annotation in *D. discoideum*.

This matters because the same QuickGO record shows that **every experimental annotation statC actually possesses concerns stress response or development**, not proliferation or defense:

GO term	Description	Evidence	Reference
GO:0006972	response to osmotic stress	IEP / IDA	PMID:17517120 , PMID:12771188
GO:0006979	response to oxidative stress	IDA	PMID:12771188
GO:0009408	response to heat	IDA	PMID:12771188
GO:0006970	response to osmotic stress	IMP	PMID:22944283
GO:1903013	response to DIF-1	HDA	PMID:25518940
GO:0031288	sorocarp morphogenesis	IMP	PMID:11336701
GO:0045892	negative regulation of transcription	IMP	—
GO:0010628	positive regulation of gene expression	IMP	PMID:24587195

The absence of any experimental evidence for the two candidate terms, contrasted against a rich set of experimental stress/development annotations, is the single most decisive observation for curation: the candidate terms are computational inferences layered onto a gene whose genuine, assayed function lies elsewhere.

F002 — *Dictyostelium* defense response is executed by TirA + NADPH oxidases in sentinel cells, not by STATc

The defense-response term can be independently refuted by identifying what actually performs anti-bacterial defense in *Dictyostelium*. Zhang et al. ([PMID:26927887](#)) demonstrated that Sentinel (S) cells of the multicellular slug produce extracellular DNA traps that trap and kill bacteria in a reactive-oxygen-species-dependent manner, and that this requires the **Toll/Interleukin-1 receptor domain-containing protein TirA** and **ROS-generating NADPH oxidases**; disruption of these genes decreased clearance of bacterial infection. Chen et al. ([PMID:17673666](#)) originally identified sentinel cells and TirA as the immune-like detoxification/defense machinery. **STATc appears in neither study**. Its only mechanistic link to oxidative conditions is as a stress-activated transcription factor (GO:0006979, IDA, [PMID:12771188](#)) — i.e., it responds to oxidative stress as a general environmental cue, which is categorically distinct from executing a *defense response* against a pathogen. Conflating "responds to oxidative stress"

with "defense response" would be a category error; the GO defense-response branch (GO:0006952) concerns responses to attack by another organism or injury, a process for which no statC evidence exists.

F003 — Canonical STAT domain architecture, but within a *Dictyostelid*-specific expansion; the proliferation IBA also draws on paralog dstB

UniProt Q54BD4 (gene *dstC*, dictyBase DDB_G0293532, 931 aa) has a bona fide STAT domain architecture: an N-terminal intrinsically disordered region carrying a DIF-inducible nuclear-import signal (aa ~6–46), a *Dictyostelium* STAT coiled-coil (Pfam PF09267), a STAT DNA-binding domain of the p53-like/RHD fold (InterPro IPR008967/IPR037059), an EF-hand (PF17901), a STATA_Ig domain (PF18214), and a C-terminal SH2 domain (aa ~802–931; PF00017). It is assigned to PANTHER family PTHR11801 (STAT1 family). So statC is a genuine STAT by sequence.

However, the three *Dictyostelium* STATs (dstA = O00910, dstB = Q70GP4, dstC = Q54BD4) form a **lineage-specific expansion**, and only the shared transcription-factor module is truly homologous to the metazoan proteins. The domains that in animals couple STATs to cytokine/interferon receptors and the JAK kinases — the functional substrate for proliferation and immune-defense signaling — have no counterpart pathway in *Dictyostelium*. statC is instead activated by a **non-canonical PTP3 phosphatase-inhibition switch** ([PMID:12771188](#), [PMID:20159963](#)): stress-induced serine phosphorylation of the tyrosine phosphatase PTP3 inhibits its activity toward STATc, allowing a semi-constitutive tyrosine kinase to phosphorylate and activate STATc. Notably, the IBA with/from set for GO:0042127 includes *Dictyostelium* DDB_G0268638 = dstB (STATb) alongside the animal STATs — direct evidence that the proliferation annotation on statC is being propagated in part from its own paralog.

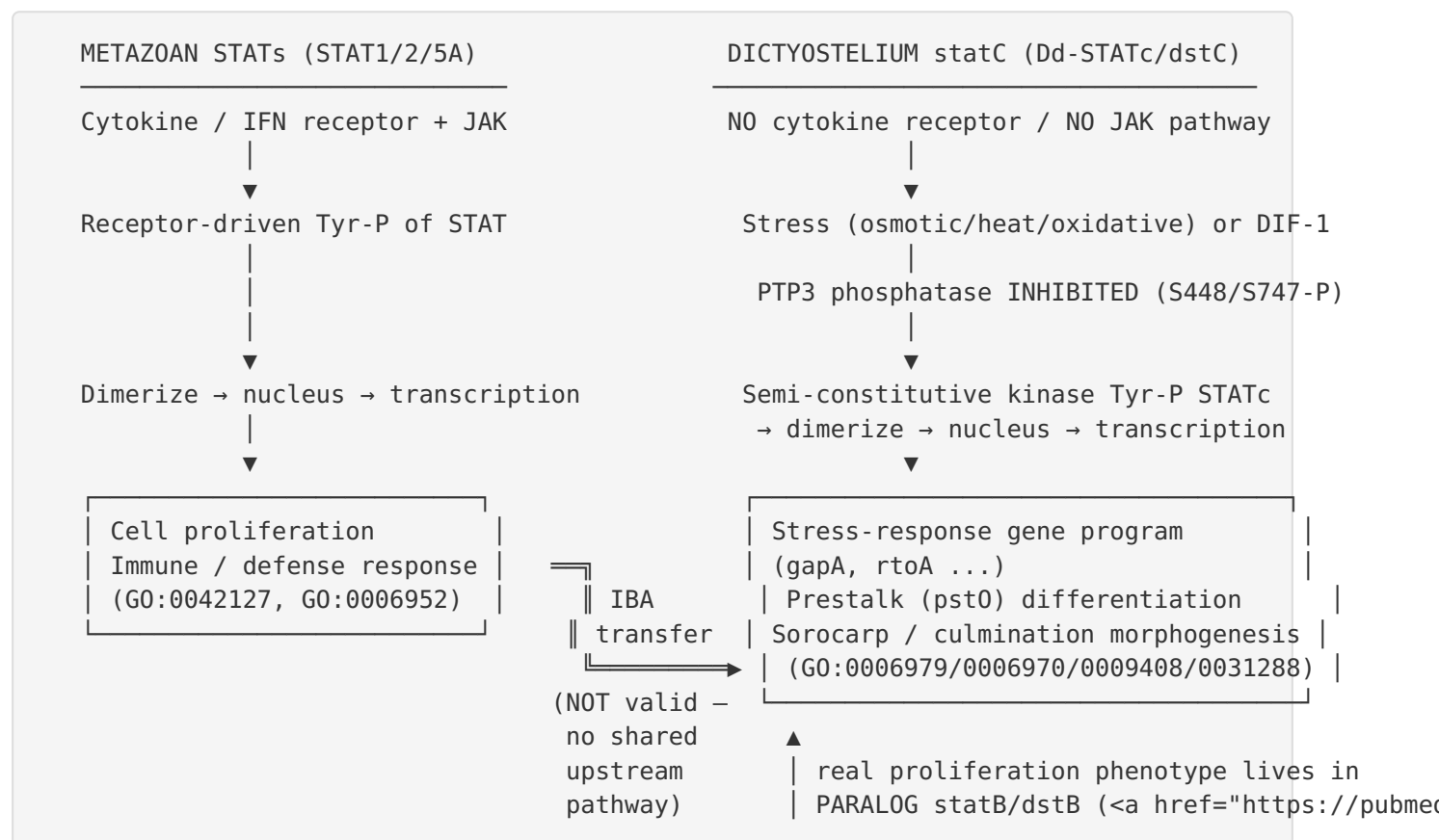
F004 — The proliferation term is paralog over-annotation from statB; defense response has no experimental support in ANY *Dictyostelium* STAT

This is the decisive discriminating finding. QuickGO shows that the paralog **dstB (STATb, Q70GP4) carries an experimental IMP annotation to GO:0042127** derived from Zhukovskaya et al. ([PMID:14701681](#)), which reports that "Dd-STATb ... has a subtle role in growth, so that Dd-STATb-null cells are gradually lost from the population when they are co-cultured with parental cells." statC (dstC) and statA carry GO:0042127 **only by IBA** (GO_REF:0000033). In other words, the one real proliferation phenotype in the *Dictyostelium* STAT family belongs to a *different* gene, and the annotation has been phylogenetically spread to its paralogs — the textbook signature of paralog over-annotation.

For GO:0006952 (defense response), all three *Dictyostelium* STATs (dstA, dstB, dstC) are annotated **only by IBA**, propagated from animal STAT1/STAT2. No experimental *Dictyostelium* defense-response annotation exists for any of them. The defense-response term therefore fails not just for statC specifically but for the entire *Dictyostelium* STAT clade.

Mechanistic Model / Interpretation

The core of the adjudication is separating **what statC directly does** from **what has been computationally attributed to it by homology**. The following model summarizes the evidence.



The immediate molecular function under test is **sequence-specific DNA-binding transcription-factor activity acting downstream of a stress/DIF-1 signal**. Everything statC does experimentally is a transcriptional response to environmental or developmental cues. The candidate proliferation and defense functions are neither immediate activities nor even downstream phenotypes documented in statC mutants — they are inferences transferred

across a homology boundary that does not carry the required upstream signaling context (cytokine receptors, JAKs, interferon response). Because the transfer crosses a mechanistic discontinuity, the resulting annotations are not merely "context-specific" or "downstream" — they are unsupported.

Adjudication of each candidate term separately, as the seed required:

Candidate term	Mechanism required in metazoa	Present in <i>Dictyostelium</i> ?	Experimental statC evidence?	Verdict
GO:0042127 regulation of cell population proliferation	Cytokine/JAK-driven growth signaling	No JAK/cytokine system	None (IBA only; real phenotype in paralog statB)	Refuted for statC — paralog carry-over
GO:0006952 defense response	IFN/immune receptor signaling	Defense done by TirA + NADPH oxidase sentinel cells	None (IBA only in all 3 STATs)	Refuted — no evidence in any Dicty STAT
(statC's real role) stress-activated TF / prestalk differentiation	—	Yes, via PTP3 switch	Multiple IDA/IMP/IEP/HDA	Supported — core function

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ limitation
QuickGO record for Q54BD4 (database)	Review/ database	Refutes (both terms)	Are GO:0042127 & GO:0006952 experimentally supported on statC?	Both terms IBA-only (ECO:0000318, GO_REF:0000033, PTN000927860); with/from = animal STATs + paralog dstB	Annotation metadata, <i>D. discoideum</i>	High for annotation provenance depends on current QuickGO
14701681	Mutant phenotype (IMP)	Competing / refutes statC proliferation	Which <i>Dictyostelium</i> STAT has a proliferation phenotype?	Dd-STATb-null cells gradually lost in co-culture with parental — subtle growth role	statB/dstB, vegetative growth	High; phenotype subtle and belongs to paralog, not statC
26927887	Mutant phenotype / direct assay	Refutes statC defense role	What executes anti-bacterial defense in <i>Dictyostelium</i> ?	Sentinel-cell DNA extracellular traps require TirA + NADPH oxidases; disruption ↓ bacterial clearance	Slug sentinel cells	High; statC implicated
17673666	Mutant phenotype	Refutes statC defense role	Identity of immune-like defense machinery	Sentinel cells + TirA provide detoxification/ immune-like function	Slug, vegetative amoebae	High; statC absent
12771188	Direct assay (IDA)	Supports core stress-TF role	Does STATc mediate stress transcriptional response?	STATc activated by osmotic/heat/ oxidative stress and DIF; drives	<i>D. discoideum</i> cells	High; establishes stress-response, defense

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence/ limitation
				gapA, rtoA induction		
17517120	Expression profiling (IEP)	Supports core stress-TF role	STATc's role in hyperosmotic response	~20% of osmotic-stress-regulated genes are STATc-dependent	Microarray, sorbitol stress	High
20159963	Direct assay	Qualifies (mechanism)	How is STATc activated?	Non-canonical activation: PTP3 inhibition via S448/S747 phosphorylation; cGMP + Ca ²⁺ second messengers	<i>D. discoideum</i>	High; con no JAK/ cytokine r
24587195	Direct assay (IMP)	Supports core stress-TF role	Upstream kinases of STATc	Pyk3/Phg2 regulate STATc phosphorylation via PTP3; STATc positively regulates its own gene set	<i>D. discoideum</i>	High
11336701	Mutant phenotype (IMP)	Supports core developmental role	STATc's developmental function	Regulates early development speed, terminal differentiation timing, ecmA prestalk expression	Developing slug	High; developm TF, not proliferat
UniProt/ InterPro/ PANTHER (Q54BD4)	Structural/ evolutionary	Qualifies	Is statC a genuine STAT?	Canonical STAT architecture (coiled-coil, DBD, SH2) but in Dictyostelid-	Sequence/ domain	High for domains; homology functional transfer

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence limitation
				specific expansion; PTHR11801		

GO Curation Implications

Lead requiring curator verification. The evidence supports the following curation actions on statC (Q54BD4):

- **GO:0042127 (regulation of cell population proliferation) — BP.** Recommend **removal or reclassification to non-core / NOT**. It is an IBA-only annotation whose with/from set includes the paralog statB/dstB, which holds the only experimental proliferation phenotype in the family. This is paralog over-annotation. If policy retains IBA calls, at minimum flag it as non-core and add a curator note that the experimental evidence is paralog-specific (statB, [P 14701681](#)).

- **GO:0006952 (defense response) — BP.** Recommend **removal or NOT**. IBA-only in all three *Dictyostelium* STATs, propagated from animal STAT1/STAT2. No experimental support in any *Dictyostelium* STAT; genuine defense is a TirA/NADPH-oxidase sentinel-cell function ([P 26927887](#)).

([P 17673666](#)).
Do not conflate with statC's oxidative-stress response (GO:0006979).

- **Retain (well-supported, core):** response to osmotic stress (GO:0006972/0006970), response to oxidative stress (GO:0006979), response to heat (GO:0009408), response to DIF-1 (GO:1903013), sorocarp morphogenesis (GO:0031288), and transcription-regulator activities (GO:0045892 negative regulation of transcription; GO:0010628 positive regulation of gene expression). These reflect the gene product's primary MF (DNA-binding transcription factor activity) and BP (stress-response transcription; prestalk differentiation).

- **MF anchor:** The most informative molecular-function term is **sequence-specific DNA-binding transcription factor activity** acting in stress/DIF-1 response — not "protein binding."

Mechanistic Scope

The immediate molecular activity being tested is that of a **latent, cytoplasmic-to-nuclear DNA-binding transcription factor**. On receipt of a stress or DIF-1 signal, statC is tyrosine-phosphorylated, dimerizes, translocates to the nucleus (via an N-terminal import signal, with DIF-regulated nuclear export control; [PMID:12506009](#)), and activates or represses target genes. That is the direct gene-product activity. Everything else attributed to statC is downstream of this transcriptional switch. The two candidate GO terms fall outside even this downstream scope:

- **Proliferation (GO:0042127)** would require statC transcription to regulate cell-cycle or growth-fitness genes in a demonstrated manner. No such statC regulon or statC-null growth phenotype is documented; the closest evidence is the paralog statB's subtle competitive-growth defect.
- **Defense response (GO:0006952)** would require statC to be an effector or transcriptional driver of an anti-pathogen program. Instead, defense is a distinct cellular module (sentinel cells, TirA, NADPH oxidases). statC's oxidative-stress responsiveness is a generic environmental-stress adaptation, not a pathogen-directed defense process.

Thus both candidate terms are separated from statC's genuine activity by at least one, and in the defense case an entire, mechanistic discontinuity.

Conflicts and Alternatives

1. **Paralog confusion (primary alternative explanation).** The strongest alternative interpretation of the proliferation annotation is that it is genuine — *for statB, not statC*. The IBA with/from set explicitly lists dstB (Q70GP4), and the experimental IMP annotation

(P 14701681)

is on statB. Any curator retaining GO:0042127 on statC should recognize this as carry-over from a paralog.

2. **Stress vs. defense category boundary.** statC is IDA-annotated to oxidative-stress response (GO:0006979). A curator could be tempted to treat this as evidence for defense response. This is an alternative that the evidence refutes: responding to oxidative stress as an environmental cue is distinct from mounting a defense response against a pathogen. The two GO branches are separate for good reason.
3. **Organism-specific signaling discontinuity.** Metazoan STAT proliferation/immune roles depend on cytokine/interferon receptors and JAK kinases, absent in *Dictyostelium*. statC is activated by a phosphatase-inhibition switch (PTP3), a mechanism unrelated to the receptor systems that drive the metazoan functions. Homology of the DNA-binding/SH2 module does not license transfer of the receptor-context-dependent functions.
4. **Database carry-over.** Both candidate terms trace to a single PANTHER node (PTN000927860). This is a systematic propagation, not independent lines of evidence, so it should be weighted as one inference, not several.

Limitations and Knowledge Gaps

- **Annotation snapshot dependency.** The provenance analysis relies on the QuickGO/UniProt annotation state at the time of retrieval. Curators should re-verify current evidence codes and with/from sets, as PANTHER-based IBA propagation is periodically re-computed.
- **No statC proliferation assay has been reported.** The gap is not that statC was tested and found negative for proliferation — it is that it appears never to have been directly assayed for a growth/proliferation phenotype in a controlled competition experiment analogous to the statB study. A statC-null co-culture competition assay would resolve whether statC has any statB-like subtle growth role.
- **No statC bacterial-clearance assay.** statC has not been tested in sentinel-cell/ET or bacterial-killing assays. Its role (if any) in defense is untested rather than experimentally excluded, though the mechanistic evidence points strongly to TirA/NADPH oxidases as the effectors.
- **Structure/PANTHER details inferred from database records,** not from a locally executed structural alignment. Domain boundaries and family assignment are as reported by UniProt/InterPro/PANTHER; independent HMM/structure verification was not run programmatically here.

- **Possible indirect roles.** Because statC broadly reprograms transcription under stress, an indirect contribution to population fitness under specific stresses cannot be formally excluded — but "indirect fitness effect under stress" is not equivalent to the specific GO terms GO:0042127 or GO:0006952.

Discriminating Tests

To most efficiently separate the seed hypothesis from the paralog-carry-over / stress-TF alternative:

1. **statC-null vs. parental co-culture competition assay** (mirroring

P 14701681

for statB). If statC-null cells are *not* progressively outcompeted, GO:0042127 on statC is refuted experimentally, not just by provenance.

2. **statC involvement in sentinel-cell extracellular traps / bacterial clearance** (mirroring

P 26927887

Assay bacterial clearance and ET formation in statC-null slugs. Expected result: no defect (defense is TirA/NADPH-oxidase driven), directly testing GO:0006952.

3. **Comparative annotation audit across the STAT1 PANTHER family** (PTN000927860) to quantify how many non-metazoan members receive proliferation/defense IBA terms — establishes whether this is systematic over-propagation warranting a family-level curation fix.

4. **ChIP-seq / RNA-seq of statC targets** under stress vs. growth to confirm that statC's regulon is dominated by stress/differentiation genes (gapA, rtoA, ecmA) rather than cell-cycle or immune-effector genes.

5. **Cross-check statB and statA** annotations: verify that statB retains its experimental GO:0042127 and that statA's GO:0042127 is likewise IBA-only, consolidating the paralog-source interpretation.

Proposed Follow-up Actions / Curation Leads

All items below are leads requiring curator verification.

- **Action change (GO:0042127 on statC):** Change from active annotation to **removed** / **non-core** / **NOT**. Rationale snippet to verify against

P 14701681

"Dd-STATb ... has a subtle role in growth, so that Dd-STATb-null cells are gradually lost from the population when they are co-cultured with parental cells." — establishes the phenotype belongs to the paralog statB, not statC.

- **Action change (GO:0006952 on statC):** Change to **removed** / **NOT**. Rationale snippets to verify:

P 26927887

"The production of ETs by S cells requires a Toll/Interleukin-1 receptor domain-containing protein TirA and reactive oxygen species-generating NADPH oxidases. Disruption of these genes results in decreased clearance of bacterial infections.";

P 17673666

"A Toll/interleukin-1 receptor (TIR) domain protein, TirA, was also required for some S cell functions." — defense executed by TirA/NADPH oxidase, not STATc.

- **Retain as core:** stress-response (GO:0006979 IDA,

P 12771188

— *"hyperosmotic stress, heat shock and oxidative stress also activate Dd-STATc"*) and developmental/prestalk transcription-factor roles (GO:0031288 IMP,

P 11336701

— *"a Dictyostelium STAT, Dd-STATc, which regulates the speed of early development and the timing of terminal differentiation"*).

- **Candidate curator note:** "GO:0042127 and GO:0006952 on statC derive from a single metazoan-STAT PANTHER IBA propagation (PTN000927860); with/from includes paralog dstB and animal STAT1/2/5A. No *Dictyostelium* experimental support for either term on any of dstA/dstB/dstC except the statB-specific IMP growth phenotype."

- **Suggested questions for curator:** (1) Does project policy retain IBA-only calls when the with/from includes an in-organism paralog with the experimental phenotype? (2) Should the defense-response term be corrected family-wide across dstA/dstB/dstC?
- **Suggested experiments:** statC-null competitive growth assay; statC-null bacterial-clearance/ET assay (see Discriminating Tests 1–2).

Bottom Line

statC is a genuine STAT-family, stress- and DIF-1-activated DNA-binding transcription factor that drives *Dictyostelium* stress-response gene expression and prestalk differentiation via a non-canonical PTP3 phosphatase switch. Its two candidate annotations — regulation of cell population proliferation and defense response — are **over-annotations**: both are IBA-only, propagated from metazoan STATs (and, for proliferation, from the paralog statB) across a signaling-context boundary (no JAK/cytokine/interferon system) that invalidates the functional transfer. The recommended lead is removal or non-core reclassification of GO:0042127 and GO:0006952 on statC, pending curator verification.

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